

Практикум E2:archive

Every past-paper question on this topic, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in Practicum E2, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

О практикуме

Тема взята целиком: 99 баллов, 26 блоков. Корпус числит 27 блоков и 105 баллов; разница — ноябрьский 2023 вопрос про $e^{(\cos 2x)}$, поставленный в TZ1 и TZ2 слово в слово. Те же две бумаги уже дали такой дубль в E1.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e2-maclaurin

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

E2 archive: Maclaurin series

Every past-paper question on this topic, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in *Practicum E2*, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

What is inside. The whole of `calculus.series`, sessions May 2021 — November 2025: **27 questions, 99 marks**, in nine sections, one section per technique.

The corpus records 27 blocks and 105 marks. One of them is a duplicate: the November 2023 question on $e^{\cos 2x}$ was set in TZ1 and in TZ2 word for word — two different papers, 8823-7106 and 8823-7111, one question. It appears here once. The count of *questions* is nevertheless 27, because the November 2023 question

is split across two sections: part (d)(i) is answered by substituting into a known series and belongs to section 2, while (d)(ii) and (d)(iii) put a series inside a series and belong to section 5.

Fourteen questions are Paper 1, four are Paper 2, eight are Paper 3 — and unlike the four topics before it, this one really does split that way. Paper 1 wants $x + x^2 + \frac{x^3}{3}$ and $m = \frac{3}{2}, -\frac{5}{2}$: exact, every time. Paper 3 wants $\pi \approx 3.1412$ and an error of 3.7×10^{-5} . The second half of this topic is arithmetic, and it is meant to be.

How to work. Read the question, answer in the cell below it, run the cell. Most of the checks here do not know the answer. `verify_maclaurin` takes the function from the question, subtracts what you wrote, and asks where the remainder starts: a Maclaurin polynomial is not some polynomial to be guessed but the one that hugs the function at zero tightly enough, and that is a property of your answer alone. `verify_series_solution` puts your polynomial into the differential

equation. `verify_terms` tries the error bound against term $n+1$ and against term n , which is the whole of the off-by-one this topic is famous for.

Six checks do compare a hash, and they are all the same case: the answer is an approximation. $\frac{61}{105}$, $\frac{500}{279}$, 3.1412, 3.7×10^{-5} — a number produced by throwing away the tail of a series has nothing to be substituted back into. E1 needed two hashes in twenty-three cells; E2 needs six, and the reason is not a weaker design but a topic whose second half is entirely about approximation.

Four questions have no cell at all. In each of them the answer is printed in the question — «show that the Maclaurin series for $\sec x$... is $1 + \frac{x^2}{2} + \frac{5x^4}{24}$ » — so there is nothing to check and everything to derive. Read the question, do the work on paper, then read the solution.

Leave a cell blank and it prints  and moves on, which means you can run the

whole notebook top to bottom on the first open and nothing breaks.

The solutions are at the end, numbered the same way. Open one after you have worked the question, not before — and read the markscheme note in it, because that is where the marks actually are.

The nine sections.

§	technique	questions	marks
1	Straight from the definition	4	13
2	Substitute into a series you know	2	2 + part of 6
3	The binomial series	1	5
4	Multiplying two series	5	21
5	A series inside a series	2	4 + part of 6
6	Differentiating and integrating a series	3	12
7	A series out of a differential equation	1	3
8	The series in place of the function	6	20
9	How wrong the approximation is	3	13

Sections 1–5 build a series and carry 45 of the 99 marks; the choice between them is half the work, and it is settled by looking at the function before writing anything. Sections 6–7 get a series out of another series or out of an equation. Sections 8–9 use it — and section 9 is the only place in the whole archive where

anyone asks how good an approximation is.

Five series are worth knowing cold, because every question here starts by recognising one of them: e^x , $\sin x$, $\cos x$, $\ln(1+x)$, $\arctan x$. They are in the formula booklet. The binomial series is there too, and the questions that need it say so.

```
import sys
sys.path.append('..')          # from
practicum/calculus to practicum/kit.py
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Rational, sqrt, pi, E, series, Sum

language('en')                 # this
notebook is in English, and so are the checks

a, b, c, d, m, n, r = symbols('a b c d m n
r')

print('ready; sympy', sp.__version__)
print('a series, written out: ', x + x**2 +
x**3/3)
print('a series with a letter:', 1 -
n*x**2/2)
print('an exact number:      ',
Rational(500, 279))
print('a decimal:           ', 3.1412)
```


1. Straight from the definition

$$\begin{aligned} f(x) &= f(0) \\ &\quad + f'(0)x \\ &\quad + \frac{f''(0)}{2!}x^2 \\ &\quad + \frac{f'''(0)}{3!}x^3 \\ &\quad + \dots \end{aligned}$$

The way in when nothing else works: differentiate, put $x = 0$, divide by $n!$. In practice the differentiating has usually been done for you in the part before — an n th derivative proved by induction, or a relation like $g'' = 2(g' - g)$ — and *hence* means you are meant to use it.

1.1 — November 2021 Paper 1 Q11(b), 3 marks

It is given that

$$\begin{aligned} \frac{d^n}{dx^n}(x^2 e^x) \\ = [x^2 + 2nx + n(n-1)] e^x, \quad n \in \mathbb{Z}^+. \end{aligned}$$

Hence or otherwise, determine the Maclaurin series of $f(x) = x^2 e^x$ in

ascending powers of x , up to and including the term in x^4 .

```
F = x**2 * exp(x)

q1_1 = ...

verify_maclaurin('1.1', q1_1, F, order=4)
```

1.2 — May 2022 TZ1 Paper 1 Q12(d), 5 marks

The function g is defined by $g(x) = e^x \cos x$. It has been shown that

$$g''(x) = 2(g'(x) - g(x))$$

and

$$g^{(4)}(x) = 2(g'''(x) - g''(x)).$$

Using this result, find the Maclaurin series for $g(x)$ up to and including the x^4 term.

```
G = exp(x) * cos(x)

q1_2 = ...

verify_maclaurin('1.2', q1_2, G, order=4)
```

1.3 — May 2025 TZ1 Paper 1 Q12(e) (i), 3 marks

Consider the family of functions $f_n(x) = \cos^n x$, where $x \in \mathbb{R}$ and $n \in \mathbb{N}$.

Find the Maclaurin series of $f_n(x)$ up to the term in x^2 .

```
FN = cos(x)**n
```

```
q1_3 = ...
```

```
verify_maclaurin('1.3', q1_3, FN, order=2,  
params={n: (2, 3, 7)})
```

1.4 — May 2024 TZ2 Paper 1 Q12(b), 2 marks

Let $f(x) = (1 - ax)^{-\frac{1}{2}}$, where $ax < 1$, $a \neq 0$.
It has been proved by induction that

$$f^{(n)}(x) = \frac{a^n (2n-1)! (1-ax)^{-\frac{2n+1}{2}}}{2^{2n-1} (n-1)!}, \quad n \in \mathbb{Z}^+.$$

By using part (a) or otherwise, show that the Maclaurin series for $f(x) = (1 - ax)^{-\frac{1}{2}}$ up to and including the x^2 term is

$$1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2.$$

This question has no cell: the answer is printed in it. The marks are in the route, and the route is in the solution.

2. Substitute into a series you know

$\sin(x^2)$, $\cos 2x$, e^{-3x} : the same five series with a different letter inside. Replace x by the whole new argument, power and all — and recount how deep the original series has to go, because $(x^2)^3 = x^6$ eats three terms of the answer in one.

2.1 — May 2024 TZ1 Paper 1 Q8(a)(i), 2 marks

Find the first two non-zero terms in the Maclaurin series of $\sin(x^2)$.

```
q2_1 = ...
```

```
verify_maclaurin('2.1', q2_1, sin(x**2),  
terms=2)
```

2.2 — November 2023 TZ1 Paper 1

Q11(d)(i), part of 6 marks

Consider the function $f(x) = e^{\cos 2x}$, where $-\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$.

Find the Maclaurin series for $\cos 2x$, up to and including the term in x^4 .

```
q2_2 = ...
```

```
verify_maclaurin('2.2', q2_2, cos(2*x),  
order=4)
```

3. The binomial series

$$\begin{aligned}(1 + u)^p &= 1 \\ &+ pu \\ &+ \frac{p(p-1)}{2!}u^2 \\ &+ \frac{p(p-1)(p-2)}{3!}u^3 \\ &+ \dots, \quad |u| < 1.\end{aligned}$$

The bracket has to start with a 1; if it does not, take the factor out first. The sign inside travels with u , and the

expansion is infinite, so the question always says where to stop.

3.1 — November 2025 TZ1 Paper 1 Q5(a), 5 marks

The first four terms of the Maclaurin series expansion of $(1 - x)^{-4}$ are

$$1 + ax + bx^2 + 20x^3, \quad a, b \in \mathbb{Z}^+.$$

(i) Show that $a = 4$. (ii) Find the value of b

Answer with the whole series up to x^3 — that is the two parts together.

```
q3_1 = ...
```

```
verify_maclaurin('3.1', q3_1, (1 - x)**-4,  
order=3)
```

4. Multiplying two series

The commonest technique in the topic: 21 of the 99 marks. Expand each factor with a little room to spare, multiply, and throw away everything above the power you were asked for **as you go**, not at the end.

One thing to watch before you start: how deep each factor has to go is not the same for both. $\sin x$ begins at x , so in $e^x \sin x$ up to x^3 the sine needs two terms and the exponential three.

4.1 — May 2022 TZ1 Paper 1 Q12(a), 4 marks

The function f is defined by $f(x) = e^x \sin x$, where $x \in \mathbb{R}$.

Find the Maclaurin series for $f(x)$ up to and including the x^3 term.

```
q4_1 = ...
```

```
verify_maclaurin('4.1', q4_1, exp(x)*sin(x),  
order=3)
```

4.2 — May 2024 TZ1 Paper 1 Q8(a) (ii), 3 marks

Find the first two non-zero terms in the Maclaurin series of $\sin^2(x^2)$.

```
q4_2 = ...
```

```
verify_maclaurin('4.2', q4_2, sin(x**2)**2,  
terms=2)
```

4.3 — May 2024 TZ1 Paper 1 Q8(b), 2 marks

Hence, or otherwise, find the first two non-zero terms in the Maclaurin series of $4x \sin(x^2) \cos(x^2)$.

```
q4_3 = ...
```

```
verify_maclaurin('4.3', q4_3,  
4*x*sin(x**2)*cos(x**2), terms=2)
```

4.4 — May 2021 TZ1 Paper 1 Q12(c), 8 marks

Let $f(x) = \sqrt{1+x}$ for $x > -1$, and let $g(x) = e^{mx}$, $m \in \mathbb{Q}$. Consider the function h defined by $h(x) = f(x) \times g(x)$ for $x > -1$.

It is given that the x^2 term in the Maclaurin series for $h(x)$ has a coefficient of $\frac{7}{4}$. Find the possible values of m .


```
H = sqrt(1 + x) * exp(m*x)
```

```
q4_4 = [...]
```

```
# The check puts each of your values back  
# where the question puts it: into  
# the coefficient of x^2, which has to come  
# out as 7/4. Then it scans the  
# window for values you missed.  
verify_roots('4.4', q4_4, series(H, x, 0,  
3).remove0().coeff(x, 2) - Rational(7, 4),  
(-10, 10), var=m)
```

4.5 — May 2024 TZ2 Paper 1 Q12(c), 4 marks

It has been shown that the Maclaurin series for $(1 - ax)^{-\frac{1}{2}}$ up to the x^2 term is $1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2$.

Hence show that

$$(1 - 2x)^{-\frac{1}{2}}(1 - 4x)^{-\frac{1}{2}} \approx \frac{2 + 6x + 19x^2}{2}.$$

This question has no cell: the answer is printed in it.

5. A series inside a series

Section 2 substituted a single power. Here what goes inside is itself a series, and it has to be raised to powers — so the arithmetic is longer and one condition matters that did not before: **the inside has to vanish at zero.** e^u is expanded about $u = 0$, and $\cos 2x$ at $x = 0$ is 1, not 0. That is why the November 2023 question walks you through $e^{\cos 2x - 1}$ first and only then puts the e back.

5.1 — *May 2021 TZ2 Paper 2 Q9(b), 4 marks*

The first three terms of the binomial expansion of $(1+t)^{-1}$ in ascending powers of t are $1 - t + t^2$.

By using the Maclaurin series for $\cos x$ and the result above, show that the Maclaurin series for $\sec x$ up to and including the term in x^4 is

$$1 + \frac{x^2}{2} + \frac{5x^4}{24}.$$

This question has no cell: the answer is printed in it.

5.2 — November 2023 TZ1 Paper 1 Q11(d)(ii) and (iii), part of 6 marks

Still with $f(x) = e^{\cos 2x}$, and using the series for $\cos 2x$ from 2.2:

(ii) Hence, find the Maclaurin series for $e^{\cos 2x - 1}$, up to and including the term in x^4 .

(iii) Hence, write down the Maclaurin series for $f(x)$, up to and including the term in x^4 .

```
q5_1 = ...           # (ii) the series for
e^(cos 2x - 1)
q5_2 = ...           # (iii) the series for
e^(cos 2x)

verify_maclaurin('5.2(ii)', q5_1,
exp(cos(2*x) - 1), order=4)
verify_maclaurin('5.2(iii)', q5_2,
exp(cos(2*x)),      order=4)
```

6. Differentiating and integrating a series

When the series you want is unknown but the series for its derivative or its integral is not. The tell is usually a geometric series in the question, because $1 + u + u^2 + \dots = \frac{1}{1-u}$ is a Maclaurin series — the one everybody uses without noticing.

Integrating brings a constant with it, and $x = 0$ is what fixes it. That is a mark on its own.

6.1 — November 2025 TZ3 Paper 3 Q2(b), 3 marks

Given $|x| < 1$, the sum to infinity of the geometric series $1 - x^2 + x^4 - x^6 + \dots$ is $\frac{1}{1+x^2}$.

Hence, use integration to show that the Maclaurin series of $\arctan x$ may be expressed as

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

This question has no cell: the answer is printed in it. It is also the foundation of every question in sections 8 and 9, so it is worth doing on paper properly.

6.2 — May 2024 TZ1 Paper 3 Q1(c)(i), 4 marks

Consider the sum of an infinite geometric sequence, with first term a and common ratio r ($|r| < 1$),

$$a + ar + ar^2 + ar^3 + \dots = \frac{a}{1 - r}.$$

By differentiating both sides of the above equation with respect to r , find an expression for $\sum_{n=1}^{\infty} n a r^{n-1}$ in terms of a and r .

```
q6_1 = ...
```

```
# The check does to the right-hand side
# exactly what the question tells you
# to do to both sides – and then compares.
verify_identity('6.2', q6_1, diff(a/(1 - r),
r), var=r)
```

**6.3 — May 2025 TZ1 Paper 2 Q12(c),
5 marks**

Consider the family of functions f_n defined by $f_n(x) = \sum_{r=0}^n (-2x^2)^r$, where $x \in \mathbb{R}$ and $n \in \mathbb{N}$.

Consider the function $f(x) = \lim_{n \rightarrow \infty} f_n(x)$, defined over the domain $-k < x < k$ where $k > 0$. The largest possible value of k is K .

(i) Find the value of K , giving your answer in exact form.

(ii) Express $f(x)$ as a rational function in the form $\frac{1}{a + bx^2}$, where a and b are constants to be determined.

```

q6_2 = ...           # (i) K, exact
q6_3 = ...           # (ii) f(x) as a rational
function

verify_exact('6.3(i)', q6_2, 1/sqrt(2))

# For (ii) the check has no closed form to
# compare against – it sums the
# series from the question far enough that
# the tail cannot be seen, and
# asks whether your rational function agrees
# inside the domain.
PARTIAL = Sum((-2*x**2)**r, (r, 0,
200)).doit()
verify_identity('6.3(ii)', q6_3, PARTIAL,
var=x,
               samples=(0.05, 0.1, 0.15,
0.2, 0.25, 0.3, 0.35, 0.4))

```

7. A series out of a differential equation

There is no formula for the function at all — only an equation and a starting value. That is enough: the equation hands you $y'(0)$, and differentiating the equation hands you the next derivative, one at a time.

The one thing that goes wrong here goes wrong every time: the right-hand side

contains y , and y is a function of x , so differentiating it produces $\frac{dy}{dx}$. Miss that and the whole tower collapses from the second derivative up.

**7.1 — May 2023 TZ1 Paper 2 Q12(c),
3 marks**

Consider the differential equation

$$\frac{dy}{dx} = \frac{x^2y - y}{x^2 + 1},$$

where $y > 0$ and $y = 3$ when $x = 0$. It has been shown that $\frac{d^2y}{dx^2} = 3$ when $x = 0$.

(i) Given that $\frac{d^3y}{dx^3} = 9$ when $x = 0$, find the first four terms of the Maclaurin series for y .

(ii) Use the Maclaurin series to find an approximate value for y when $x = 0.15$. Give your answer correct to six significant figures.


```

RHS = (x**2*y - y)/(x**2 + 1)

q7_1 = ...           # (i)  the first four
terms
q7_2 = ...           # (ii) y(0.15) to six
significant figures

# No function to compare with – there is no
formula for y. The check puts
# your polynomial into the equation instead,
and into the initial condition.
verify_series_solution('7.1(i)', q7_1, RHS,
ic=3, order=3)
check_num('7.1(ii)', q7_2, 6, '98ffa9c9a889')

```

8. The series in place of the function

The series is in hand — usually from the part just above, and *hence* says so. Now the polynomial stands in for the function, and you do to it what you could not do to her: cancel a power and read off a limit, substitute a number, integrate.

$\int_0^1 e^{x^2} \sin(x^2) dx$ has no antiderivative in closed form. That is not an obstacle to the question — it is the question.

8.1 — May 2022 TZ1 Paper 1 Q12(b), 4 marks

The Maclaurin series for $e^x \sin x$ up to the x^3 term is $x + x^2 + \frac{x^3}{3}$ (question 4.1).

Hence, find an approximate value for

$$\int_0^1 e^{x^2} \sin(x^2) dx.$$

```
q8_1 = ... # exact fraction  
  
check_expr('8.1', q8_1, '3f21f8467409')
```

8.2 — May 2024 TZ2 Paper 1 Q12(e), 5 marks

It has been shown that $(1 - 2x)^{-\frac{1}{2}}(1 - 4x)^{-\frac{1}{2}} \approx \frac{2+6x+19x^2}{2}$, valid for $|x| < \frac{1}{4}$.

Use $x = \frac{1}{10}$ to determine an approximate value for $\sqrt{3}$. Give your answer in the form $\frac{c}{d}$, where $c, d \in \mathbb{Z}^+$.

```
q8_2 = ...  
  
check_expr('8.2', q8_2, '23f65acfbf92')
```

8.3 — November 2025 TZ3 Paper 3 Q2(c), 3 marks

Using $x = \frac{1}{\sqrt{3}}$ and the first **three** (non-zero) terms of the Maclaurin series of

$\arctan x$, find an approximation for π to three decimal places.

```
q8_3 = ...
```

```
check_num('8.3', q8_3, 4, '136a581f994e')
```

8.4 — November 2025 TZ3 Paper 3 Q2(g), 2 marks

It has been shown by parts that

$$\int_0^{\frac{1}{\sqrt{3}}} \arctan x \, dx = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3},$$

and the same integral, taken over the first four terms of the series for $\arctan x$, comes to 0.158422 (to six decimal places).

Use these two results to find an approximation for π . Give your answer to four decimal places.

```
q8_4 = ...
```

```
check_num('8.4', q8_4, 5, '5f0bc27eb9d3')
```

8.5 — May 2021 TZ1 Paper 3 Q2(e)(i), 3 marks

The Maclaurin series for $\tan x$ is $x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$

Use it to find $\lim_{n \rightarrow \infty} \left(4n \tan \frac{\pi}{n} \right)$.

```
q8_5 = ...
```

```
verify_limit('8.5', q8_5, 4*n*tan(pi/n),  
var=n, point=oo)
```

8.6 — May 2021 TZ2 Paper 2 Q9(c), 3 marks

By using the Maclaurin series for $\arctan x$ and the series for $\sec x$ from 5.1, find

$$\lim_{x \rightarrow 0} \left(\frac{x \arctan 2x}{\sec x - 1} \right).$$

```
q8_6 = ...
```

```
verify_limit('8.6', q8_6,  
x*arctan(2*x)/(sec(x) - 1))
```

9. How wrong the approximation is

The only place in the archive where the question is not *what does the series give* but *how far off is it*. It works because the series for $\arctan x$ alternates and its terms shrink:

Theorem. For alternating series with terms of decreasing magnitude, the error obtained in using a finite number of terms is less than or equal to the absolute value of the next term in the sequence.

All three questions below turn on the same sentence, and so does the one mistake everyone makes: the error from n terms is bounded by term number $n + 1$.

9.1 — November 2025 TZ3 Paper 3 Q2(d), 3 marks

Determine how many (non-zero) terms of the series would need to be used, such that the error in approximating $\arctan\left(\frac{1}{\sqrt{3}}\right)$ is less than 0.0001.

```
# Term number k of the series for
arctan(1/sqrt(3)), ignoring its sign.
TERM = (1/sqrt(3))**(2*k - 1) / (2*k - 1)

q9_1 = ...

verify_terms('9.1', q9_1, TERM, 0.0001)
```

9.2 — November 2025 TZ3 Paper 3

Q2(h), 5 marks

$$\int_0^{\frac{1}{\sqrt{3}}} \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right) dx$$

may be considered as the sum of alternating terms, and $\int_0^{1/\sqrt{3}} \arctan x \, dx$ is approximated using the sum of the **first four** definite integrals — which comes to 0.158422.

The exact value of the integral is $\frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3}$.

Verify that the theorem holds in this case: state the actual error, and state the bound the theorem gives.

```
q9_2 = ...           # the actual error, to
two significant figures
q9_3 = ...           # the bound the theorem
gives, exactly

check_num('9.2 the error', q9_2, 2,
'8f850b1e362c')
check_num('9.2 the bound', q9_3, 3,
'b8b1bd60ff84')
```

9.3 — November 2025 TZ3 Paper 3

Q2(i), 5 marks

Suppose that the maximum error in approximating $\int_0^{1/\sqrt{3}} \arctan x \, dx$ is required to be at most 1×10^{-6} .

Determine the smallest number of (non-zero) terms of the Maclaurin series for $\arctan x$ that should be used.

```
# Now the terms being added are integrals,
not values: term number k is
# the integral of x^(2k-1)/(2k-1) from 0 to
1/sqrt(3).
INTEGRAL = integrate(x**(2*k - 1)/(2*k - 1),
(x, 0, 1/sqrt(3)))

q9_4 = ...

verify_terms('9.3', q9_4, INTEGRAL, 1e-6,
strict=False)
```



Solutions

Numbered as above. Read one after you have worked the question.

1.1 — November 2021 P1 Q11(b), 3 marks

The given formula hands over every derivative at once. Put $x = 0$:

$$f^{(n)}(0) = [0 + 0 + n(n-1)]e^0 = n(n-1),$$

so $f(0) = 0$, $f'(0) = 0$, $f''(0) = 2$, $f'''(0) = 6$, $f^{(4)}(0) = 12$, and

$$\begin{aligned} f(x) &= \frac{2}{2!}x^2 \\ &+ \frac{6}{3!}x^3 \\ &+ \frac{12}{4!}x^4 \\ &+ \dots \\ &= x^2 + x^3 + \frac{x^4}{2} + \dots \end{aligned}$$

The formula is stated for $n \in \mathbb{Z}^+$, so $f(0)$ has to come from f itself; it is 0, which is why the series starts at x^2 .

Otherwise: multiply x^2 by the series for e^x and read off three terms. That is section 4's technique, it is faster here, and the markscheme allows it — but the question is placed after an induction proof for a reason, and «hence» is where the marks are.

1.2 — May 2022 TZ1 P1 Q12(d), 5 marks

Substituting $x = 0$ turns the two relations into arithmetic. With $g(x) = e^x \cos x$: $g(0) = 1$ and $g'(0) = 1$ (product rule, or $g' = e^x(\cos x - \sin x)$). Then

$$\begin{aligned}g''(0) &= 2(g'(0) - g(0)) = 2(1 - 1) = 0, \\g'''(0) &= 2(g''(0) - g'(0)) = 2(0 - 1) = -2, \\g^{(4)}(0) &= 2(g'''(0) - g''(0)) \\&= 2(-2 - 0) \\&= -4,\end{aligned}$$

so

$$\begin{aligned}
g(x) &= 1 \\
&+ x \\
&+ \frac{0}{2!}x^2 \\
&- \frac{2}{3!}x^3 \\
&- \frac{4}{4!}x^4 \\
&+ \dots \\
&= 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \dots
\end{aligned}$$

The relation $g''' = 2(g'' - g')$ used in the middle line is not given — it is the same differentiation done once more, and it is expected.

Note the missing x^2 : the coefficient really is zero, and an answer that quietly renumbers the terms to avoid the gap is wrong.

1.3 — May 2025 TZ1 P1 Q12(e)(i), 3 marks

$f_n(x) = \cos^n x$, so $f_n(0) = 1$ and $f'_n(x) = -n \cos^{n-1} x \sin x$, which is 0 at $x = 0$. Then

$$f_n''(x) = n(n-1) \cos^{n-2} x \sin^2 x - n \cos^n x,$$

$$f_n''(0) = -n,$$

and

$$f_n(x) = 1 - \frac{n}{2}x^2 + \dots$$

Otherwise, and much faster: $\cos x = 1 - \frac{x^2}{2} + \dots$, so $\cos^n x = \left(1 - \frac{x^2}{2} + \dots\right)^n$, and the binomial expansion keeps only $1 + n\left(-\frac{x^2}{2}\right)$ up to x^2 . That is section 3 doing section 1's work.

The next part of the paper wants $\lim_{x \rightarrow 0} \frac{f_n(x) - 1}{x^2}$, which is now $-\frac{n}{2}$ by inspection — that part belongs to E1.

1.4 — May 2024 TZ2 P1 Q12(b), 2 marks

From the proved formula, at $x = 0$,

$$f^{(n)}(0) = \frac{a^n(2n-1)!}{2^{2n-1}(n-1)!}.$$

So $f'(0) = \frac{a \cdot 1!}{2 \cdot 0!} = \frac{a}{2}$ and $f''(0) = \frac{a^2 \cdot 3!}{2^3 \cdot 1!} = \frac{6a^2}{8} = \frac{3a^2}{4}$, and

$$\begin{aligned}
f(x) &= 1 \\
&+ \frac{a}{2}x \\
&+ \frac{3a^2}{4} \cdot \frac{x^2}{2!} \\
&= 1 \\
&+ \frac{1}{2}ax \\
&+ \frac{3}{8}a^2x^2. \quad \mathbf{AG}
\end{aligned}$$

$f(0) = 1$ comes from f , not from the formula — the formula starts at $n = 1$. Two marks, and both of them are the division by $n!$: the numbers $\frac{a}{2}$ and $\frac{3a^2}{4}$ are $f'(0)$ and $f''(0)$, not the coefficients.

Otherwise: the binomial series with $p = -\frac{1}{2}$ and $u = -ax$ gives the same three terms in one line. The markscheme allows it, and part (a) is what makes the long way worth the marks.

2.1 — May 2024 TZ1 P1 Q8(a)(i), 2 marks

$\sin u = u - \frac{u^3}{3!} + \dots$ with $u = x^2$:

$$\sin(x^2) = x^2 - \frac{(x^2)^3}{3!} + \dots = x^2 - \frac{x^6}{6} + \dots$$

A1 for each term. The whole of the question is $(x^2)^3 = x^6$, and the standard slip is $-\frac{x^3}{6}$ — cubing the exponent instead of the argument.

The markscheme note that governs all of question 8: *condone presence of any additional terms once the first two correct terms are seen.*

2.2 — November 2023 TZ1 P1 Q11(d) (i), part of 6 marks

$$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \dots \text{ with } u = 2x:$$

$$\begin{aligned} \cos 2x &= 1 \\ &\quad - \frac{4x^2}{2} \\ &\quad + \frac{16x^4}{24} \\ &\quad - \dots \\ &= 1 - 2x^2 + \frac{2x^4}{3} - \dots \end{aligned}$$

Both the 4 and the 16 come from the substitution, and both are where the

marks go. Three terms of the cosine series are needed to reach x^4 , not two.

3.1 — November 2025 TZ1 P1 Q5(a), 5 marks

(i) With $p = -4$ and $u = -x$: the coefficient of x is $p \cdot (-1) = 4$, so $a = 4$. **AG**

The markscheme is firm about this: *do not award the mark for $1 + 4x$ seen without clear evidence of a substitution.* Writing down the answer to an **AG** earns nothing.

(ii) The next coefficient is

$$\frac{p(p-1)}{2!}(-x)^2 = \frac{(-4)(-5)}{2}x^2 = 10x^2,$$

so $b = 10$. Check against the given $20x^3$: $\frac{(-4)(-5)(-6)}{6}(-x)^3 = 20x^3$ — the three minus signs and the one from $(-x)^3$ cancel, which is why every coefficient of $(1-x)^{-4}$ is positive.

Otherwise, by differentiation: $f'(x) = 4(1-x)^{-5}$ gives $f'(0) = 4$; $f''(x) = 20(1-x)^{-6}$ gives $f''(0) = 20$, and $b = \frac{20}{2!} = 10$.

4.1 — May 2022 TZ1 P1 Q12(a), 4 marks

$$\begin{aligned}
 e^x &= 1 \\
 &+ x \\
 &+ \frac{x^2}{2} \\
 &+ \frac{x^3}{6} \\
 &+ \dots, \quad \sin x \\
 &= x - \frac{x^3}{6} + \dots
 \end{aligned}$$

Multiply, keeping powers up to x^3 :

$$\begin{aligned}
 e^x \sin x &= x \\
 &+ x^2 \\
 &+ \left(\frac{1}{2} - \frac{1}{6} \right) x^3 \\
 &+ \dots \\
 &= x + x^2 + \frac{x^3}{3} + \dots
 \end{aligned}$$

M1 for recognising both series, **M1** for the attempt to multiply up to x^3 , **A1A1** for the working and the answer.

Note the asymmetry: the exponential needs three terms and the sine two,

because the sine starts at x and every one of its terms is pushed up a power by the multiplication.

Otherwise, the derivatives: $f' = e^x(\cos x + \sin x)$, $f'' = 2e^x \cos x$, $f''' = 2e^x(\cos x - \sin x)$, giving 0, 1, 2, 2 at zero — the same series in four times the work.

4.2 — May 2024 TZ1 P1 Q8(a)(ii), 3 marks

Square the answer to 2.1:

$$\begin{aligned}\sin^2(x^2) &= \left(x^2 - \frac{x^6}{6} + \dots\right)^2 \\ &= x^4 - 2 \cdot x^2 \cdot \frac{x^6}{6} + \dots \\ &= x^4 - \frac{x^8}{3} + \dots\end{aligned}$$

The middle term is the whole question, and the markscheme says so with an unusual **M0**: *award M0 for $(x^2)^2 - \left(\frac{x^6}{3!}\right)^2$* — squaring term by term.

Otherwise: $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ with $\theta = x^2$, then the cosine series. Same answer, and

no cross term to forget.

4.3 — May 2024 TZ1 P1 Q8(b), 2 marks

$4x \sin(x^2) \cos(x^2) = 2x \cdot 2 \sin(x^2) \cos(x^2) = 2x \sin(2x^2)$, and

$$\begin{aligned} 2x \sin(2x^2) &= 2x \left(2x^2 - \frac{(2x^2)^3}{6} + \dots \right) \\ &= 4x^3 - \frac{8x^7}{3} + \dots \end{aligned}$$

Otherwise, and this is what «hence» means here: differentiate 4.2. $\frac{d}{dx} \sin^2(x^2) = 4x \sin(x^2) \cos(x^2)$, and differentiating $x^4 - \frac{x^8}{3}$ term by term gives $4x^3 - \frac{8x^7}{3}$ in one line. Two marks, thirty seconds.

That is section 6's technique arriving early, and it is the reason parts (a)(ii) and (b) are next to each other.

4.4 — May 2021 TZ1 P1 Q12(c), 8 marks

$$\begin{aligned}
 f(x) &= \sqrt{1+x} \\
 &= 1 \\
 &\quad + \frac{x}{2} \\
 &\quad - \frac{x^2}{8} \\
 &\quad + \dots, \quad g(x) \\
 &= e^{mx} \\
 &= 1 + mx + \frac{m^2 x^2}{2} + \dots
 \end{aligned}$$

The x^2 coefficient of the product is

$$\frac{m^2}{2} + \frac{1}{2} \cdot m + 1 \cdot \left(-\frac{1}{8}\right) = \frac{m^2}{2} + \frac{m}{2} - \frac{1}{8}.$$

Set it equal to $\frac{7}{4}$:

$$\begin{aligned}
 &\frac{m^2}{2} \\
 &\quad + \frac{m}{2} \\
 &\quad - \frac{1}{8} \qquad \qquad \qquad = \frac{7}{4} \\
 \implies &4m^2 + 4m - 15 \\
 &= 0 \\
 \implies &(2m + 5)(2m - 3) \\
 &= 0,
 \end{aligned}$$

so $m = \frac{3}{2}$ or $m = -\frac{5}{2}$.

Eight marks, and the arithmetic of the product is only three of them. The markscheme's method 3 goes the long way — $h''(0) = f(0)g''(0) + 2f'(0)g'(0) + f''(0)g(0)$ — and lands on $m^2 + m - \frac{1}{4} = \frac{7}{2}$, the same quadratic after doubling. Both routes are worth the same; the series route is shorter, because f 's expansion was already needed in parts (a) and (b).

The trap is the factor $2!$. The x^2 *coefficient* is $\frac{h''(0)}{2!}$, and confusing the two turns $\frac{7}{4}$ into $\frac{7}{2}$ and the answer into nonsense.

4.5 — May 2024 TZ2 P1 Q12(c), 4 marks

Take $a = 2$ and $a = 4$ in the result of 1.4:

$$\begin{aligned}(1 - 2x)^{-\frac{1}{2}} &\approx 1 \\ &+ x \\ &+ \frac{3}{2}x^2, \quad (1 - 4x)^{-\frac{1}{2}} \\ &\approx 1 + 2x + 6x^2.\end{aligned}$$

Multiply, up to x^2 :

$$\begin{aligned} &1 \\ &+ 3x \\ &+ \left(6 + 2 + \frac{3}{2}\right)x^2 &&= 1 \\ &+ 3x \\ &+ \frac{19}{2}x^2 \\ &= \frac{2 + 6x + 19x^2}{2}. &&\mathbf{AG} \end{aligned}$$

The middle coefficient is $6 + 2 + \frac{3}{2}$: the x^2 of one, the $x \cdot x$ cross term, and the x^2 of the other. Dropping the cross term is the standard loss, and it gives $\frac{15}{2}$.

The form matters. **AG** means the answer as printed, over 2; an answer left as $1 + 3x + \frac{19}{2}x^2$ is the same number and not the same statement.

5.1 — May 2021 TZ2 P2 Q9(b), 4 marks

$$\begin{aligned}\sec x &= \frac{1}{\cos x} \\ &= \frac{1}{1 - (1 - \cos x)} \\ &= (1 + t)^{-1}, \quad t \\ &= \cos x - 1.\end{aligned}$$

From the cosine series, $t = -\frac{x^2}{2} + \frac{x^4}{24} - \dots$,
and from part (a), $(1 + t)^{-1} = 1 - t + t^2 - \dots$
:

$$\begin{aligned}
& \sec x &= 1 \\
& - \left(-\frac{x^2}{2} + \frac{x^4}{24} \right) \\
& + \left(-\frac{x^2}{2} \right)^2 \\
& + \dots \\
& = 1 \\
& + \frac{x^2}{2} \\
& - \frac{x^4}{24} \\
& + \frac{x^4}{4} \\
& + \dots \\
& = 1 \\
& + \frac{x^2}{2} \\
& + \frac{5x^4}{24}. \quad \mathbf{AG}
\end{aligned}$$

The $\frac{x^4}{4}$ comes from t^2 , and it is the mark. Only the $-\frac{x^2}{2}$ part of t matters inside t^2 ; the $\frac{x^4}{24}$ squared is x^8 and out of range.

Everything rests on writing $\cos x$ as $1 + t$ with $t \rightarrow 0$. The expansion of $(1 + t)^{-1}$ is valid for $|t| < 1$, and $t = \cos x - 1$ is near zero when x is — which is exactly why

the question hands you $(1+t)^{-1}$ in part (a) rather than letting you loose on $\frac{1}{\cos x}$.

Otherwise: four derivatives of $\sec x$. It works and nobody does it twice.

5.2 — November 2023 TZ1 P1 Q11(d) (ii) and (iii), part of 6 marks

(ii) With $u = \cos 2x - 1 = -2x^2 + \frac{2x^4}{3} - \dots$ (from 2.2), which does vanish at $x = 0$:

$$\begin{aligned}e^u &= 1 + u + \frac{u^2}{2} + \dots \\&= 1 \\&\quad + \left(-2x^2 + \frac{2x^4}{3}\right) \\&\quad + \frac{(-2x^2)^2}{2} \\&\quad + \dots \\&= 1 - 2x^2 + \frac{2x^4}{3} + 2x^4 + \dots \\&= 1 - 2x^2 + \frac{8x^4}{3} + \dots\end{aligned}$$

(iii) $e^{\cos 2x} = e \cdot e^{\cos 2x - 1}$, so

$$f(x) = e \left(1 - 2x^2 + \frac{8x^4}{3} \right) + \dots$$

Write down — one mark, no working expected, and the whole of it is not losing the e.

Why the question is built this way: $e^{\cos 2x}$ cannot be expanded by substituting into e^u directly, because at $x = 0$ the exponent is 1, not 0, and the series for e^u is about $u = 0$. Subtracting the 1 and putting the factor e back outside is the only honest route, and part (ii) is there to force it.

The x^4 coefficient collects two contributions, $\frac{2}{3}$ from u and 2 from $\frac{u^2}{2}$. Taking only the first term of u inside u^2 is fine here — $(-2x^2)^2 = 4x^4$ and the cross term is x^6 — but taking only the first term of u *outside* loses $\frac{2}{3}$.

6.1 — November 2025 TZ3 P3 Q2(b), 3 marks

$\frac{1}{1+x^2}$ is the sum of the geometric series, and it is also $\frac{d}{dx} \arctan x$. So integrate both sides term by term:

$$\begin{aligned}
 \arctan x &= \int (1 - x^2 + x^4 - x^6 + \dots) dx \\
 &= x \\
 &\quad - \frac{x^3}{3} \\
 &\quad + \frac{x^5}{5} \\
 &\quad - \frac{x^7}{7} \\
 &\quad + \dots \\
 &\quad + C.
 \end{aligned}$$

Putting $x = 0$: $\arctan 0 = 0 = C$, so $C = 0$.

AG

Three marks: the recognition that the sum is $\arctan'$, the term-by-term integration, and the constant. The constant is a separate mark and it is the one people skip — « $+C$, and $C = 0$ since $\arctan 0 = 0$ » is a full line of the markscheme.

Everything in sections 8 and 9 is this series, so it is worth being able to produce it rather than recall it.

6.2 — May 2024 TZ1 P3 Q1(c)(i), 4 marks

$$\begin{aligned} \frac{d}{dr} (a + ar + ar^2 + ar^3 + \dots) &= a \\ &+ 2ar \\ &+ 3ar^2 \\ &+ \dots = \sum_{n=1}^{\infty} n \end{aligned}$$

$$a r^{n-1},$$

$$\frac{d}{dr} \left(\frac{a}{1-r} \right) = \frac{a}{(1-r)^2},$$

$$\text{so } \sum_{n=1}^{\infty} n a r^{n-1} = \frac{a}{(1-r)^2}.$$

The right-hand side is $a(1-r)^{-1}$, and the chain rule contributes the -1 from differentiating $(1-r)$, which cancels the -1 from the power. The square, and the plus sign, both come from there.

a is a constant here — it is the first term, not a function of r — so it rides along untouched. The next part of the question sets $a = p$ and $r = 1 - p$ and gets $E(X) = \frac{1}{p}$; that part is statistics.

6.3 — May 2025 TZ1 P2 Q12(c), 5 marks

(i) $f_n(x) = \sum_{r=0}^n (-2x^2)^r$ is geometric with ratio $-2x^2$. The limit as $n \rightarrow \infty$ exists exactly when $|-2x^2| < 1$, that is $x^2 < \frac{1}{2}$, that is $|x| < \frac{1}{\sqrt{2}}$. So

$$K = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

Exact form is a requirement on the writing: 0.707 earns nothing.

(ii) Inside that domain the sum to infinity is

$$f(x) = \frac{1}{1 - (-2x^2)} = \frac{1}{1 + 2x^2},$$

so $a = 1$ and $b = 2$.

This is section 6 in reverse. Everywhere else a geometric series is used to *build* a Maclaurin series; here a Maclaurin series is recognised as geometric and folded back into the function it came from. Both directions are the single identity $\frac{1}{1-u} =$

$1 + u + u^2 + \dots$, and knowing it in both directions is what the question is testing.

7.1 — May 2023 TZ1 P2 Q12(c), 3 marks

(i) From the given values: $y(0) = 3$; the equation at $x = 0$ gives

$$\left. \frac{dy}{dx} \right|_0 = \frac{0 - 3}{0 + 1} = -3;$$

and $y''(0) = 3$, $y'''(0) = 9$ are given. So

$$\begin{aligned} y &= 3 \\ &\quad - 3x \\ &\quad + \frac{3}{2!}x^2 \\ &\quad + \frac{9}{3!}x^3 \\ &\quad + \dots \\ &= 3 - 3x + \frac{3}{2}x^2 + \frac{3}{2}x^3 + \dots \end{aligned}$$

Both $\frac{3}{2}$'s are the division by a factorial, and both are marks: $\frac{3}{2!} = \frac{3}{2}$ and $\frac{9}{3!} = \frac{3}{2}$. Writing $3x^2 + 9x^3$ is the standard loss.

(ii) At $x = 0.15$:

$$y \approx 3$$

$$- 0.45$$

$$+ \frac{3}{2}(0.0225)$$

$$+ \frac{3}{2}(0.003375) = 2.588$$

$$8125,$$

so 2.58881 to six significant figures.

Six significant figures is the point of the question. The same $y(0.15)$ is asked three times in this paper — by Euler's method in (a), by this series, and from the exact solution in (d) — and the three answers differ in the fourth figure. Rounding early destroys the comparison the question was built for.

8.1 — May 2022 TZ1 P1 Q12(b), 4 marks

Substituting x^2 for x in the series of 4.1:

$$e^{x^2} \sin(x^2) = x^2 + x^4 + \frac{x^6}{3} + \dots$$

Integrate term by term:

$$\begin{aligned}
 \int_0^1 \left(x^2 + x^4 + \frac{x^6}{3} \right) dx &= \left[\frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{21} \right]_0^1 \\
 &= \frac{1}{3} + \frac{1}{5} + \frac{1}{21} \\
 &= \frac{61}{105}.
 \end{aligned}$$

$e^{x^2} \sin(x^2)$ has no antiderivative in closed form. That is not a difficulty in the question — it is the reason for the question, and the reason the word is *approximate*.

A1 for the substituted series, **M1** for substituting and integrating, **A1** for the integrated form, **A1** for $\frac{61}{105}$. The markscheme note: *condone absence of limits up to this stage* — the limits only have to appear when you evaluate.

8.2 — May 2024 TZ2 P1 Q12(e), 5 marks

At $x = \frac{1}{10}$ the approximation gives

$$\frac{2 + 0.6 + 0.19}{2} = \frac{2.79}{2} = \frac{279}{200},$$

and the exact left-hand side is

$$\left(\frac{4}{5}\right)^{-\frac{1}{2}} \left(\frac{3}{5}\right)^{-\frac{1}{2}} = \frac{1}{\sqrt{0.48}} = \frac{10}{\sqrt{48}} = \frac{10}{4\sqrt{3}}.$$

So

$$\begin{aligned} \frac{10}{4\sqrt{3}} &\approx \frac{279}{200} \\ &\Rightarrow \frac{1}{\sqrt{3}} \\ &\approx \frac{279}{500} \\ &\Rightarrow \sqrt{3} \\ &\approx \frac{500}{279}. \end{aligned}$$

Worth stopping on. The same statement rearranged differently gives a different answer. Writing $\frac{10}{4\sqrt{3}}$ as $\frac{5\sqrt{3}}{6}$ first and *then* solving gives

$$\sqrt{3} \approx \frac{6}{5} \cdot \frac{279}{200} = \frac{837}{500} = 1.674,$$

against $\frac{500}{279} = 1.7921$ from the route above. The true value is 1.7321, so the two answers straddle it: one is 3.5% high, the other 3.4% low. Neither is nonsense and neither is better — they are the same error pointing in opposite directions.

That is exactly what an approximate equality does when you rearrange it. The approximation $\frac{279}{200}$ undershoots the exact $\frac{10}{4\sqrt{3}}$ by a factor of 0.9665. If $\sqrt{3}$ sits in the denominator when you isolate it, it inherits $\frac{1}{0.9665}$; if you have moved it to the numerator first, it inherits 0.9665. An equation survives being multiplied through by $\sqrt{3}$; an *approximate* equation carries its error across with it.

Only one of the two is the markscheme's, and the rule that produces it is worth keeping: leave $\sqrt{3}$ where the exact side puts it, and isolate it on the last line.

The approximation is crude to begin with — 1.395 against a true 1.4434 — because $x = \frac{1}{10}$ is not that small next to the radius $\frac{1}{4}$, and only three terms were kept. A 3.4% error in the product is a 3.4% error in $\sqrt{3}$, whichever way round you write it.

8.3 — November 2025 TZ3 P3 Q2(c), 3 marks

$\arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$, so $\pi = 6 \arctan \frac{1}{\sqrt{3}}$. With three terms,

$$\begin{aligned}\arctan \frac{1}{\sqrt{3}} &\approx \frac{1}{\sqrt{3}} \\ &\quad - \frac{1}{3} \left(\frac{1}{\sqrt{3}} \right)^3 \\ &\quad + \frac{1}{5} \left(\frac{1}{\sqrt{3}} \right)^5 \\ &= 0.526\,030\dots\end{aligned}$$

and $\pi \approx 6 \times 0.526030 = 3.156$ to three decimal places.

The recognition that $\arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$ is the first mark and the whole idea: the series computes a number that happens to be a known fraction of π , and that is how π gets approximated at all.

Three terms is not many — 3.156 against 3.14159 — which is what makes parts (d) to (i) worth asking.

8.4 — November 2025 TZ3 P3 Q2(g), 2 marks

Set the exact value equal to the approximation:

$$\frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3} \approx 0.158422.$$

$\frac{1}{2} \ln \frac{4}{3} = 0.143841\dots$, so

$$\begin{aligned}\frac{\pi}{6\sqrt{3}} &\approx 0.302263\dots, & \pi \\ &\approx 6\sqrt{3} \times 0.302263 \\ &= 3.1412.\end{aligned}$$

Better than 8.3 by two decimal places, from the same series — because integrating a series smooths it: each term is divided by one more power, and the tail shrinks faster.

Four decimal places is asked for, so carry six through the arithmetic. The approximation 0.158422 is itself given to six.

8.5 — May 2021 TZ1 P3 Q2(e)(i), 3 marks

Put $u = \frac{\pi}{n}$, which $\rightarrow 0$ as $n \rightarrow \infty$. With $\tan u = u + \frac{u^3}{3} + \dots$,

$$\begin{aligned}4n \tan \frac{\pi}{n} &= 4n \left(\frac{\pi}{n} + \frac{\pi^3}{3n^3} + \dots \right) \\&= 4\pi \\&\quad + \frac{4\pi^3}{3n^2} \\&\quad + \dots \rightarrow 4\pi.\end{aligned}$$

Every term after the first carries $\frac{1}{n^2}$ or smaller, so they all vanish and the limit is the first term alone.

In the question $4n \tan \frac{\pi}{n}$ is the common value of the area and the perimeter of a regular n -gon with $A = P$; part (e)(ii) asks what 4π means, and the answer is the circle — area and circumference of a circle of radius 2 are both 4π .

8.6 — May 2021 TZ2 P2 Q9(c), 3 marks

$\arctan 2x = 2x - \frac{(2x)^3}{3} + \dots$ and, from 5.1, $\sec x - 1 = \frac{x^2}{2} + \frac{5x^4}{24} + \dots$, so

$$\begin{aligned}
\frac{x \arctan 2x}{\sec x - 1} &= \frac{2x^2 - \frac{8x^4}{3} + \dots}{\frac{x^2}{2} + \frac{5x^4}{24} + \dots} \\
&= \frac{2 - \frac{8x^2}{3} + \dots}{\frac{1}{2} + \frac{5x^2}{24} + \dots} \longrightarrow \frac{2}{\frac{1}{2}} \\
&= 4.
\end{aligned}$$

Cancel x^2 from top and bottom and the indeterminacy is gone; what is left is a ratio of constant terms.

This is a $\frac{0}{0}$ limit that l'Hôpital would also settle — twice over, with a second derivative of $\sec x$ on the way. The series route is three lines because both series were already built in parts (a) and (b), and «*by using...*» says to use them.

9.1 — November 2025 TZ3 P3 Q2(d), 3 marks

Using n terms leaves out term $n + 1$ first, and by the theorem the error is at most that term. With $x = \frac{1}{\sqrt{3}}$, term number m has magnitude

$$\frac{1}{2m-1} \left(\frac{1}{\sqrt{3}} \right)^{2m-1}.$$

Evaluate:

$$m = 4 : 0.003054,$$

$$m = 5 : 0.000792,$$

$$m = 6 : 0.000216,$$

$$m = 7 : 0.000061.$$

The first one below 0.0001 is $m = 7$, so the last term **kept** is number 6: **6 non-zero terms**.

The markscheme allows solving $\frac{1}{2m-1} \left(\frac{1}{\sqrt{3}}\right)^{2m-1} < 0.0001$ on the GDC, which gives $m = 6.60\dots$, hence $m = 7$ for the first omitted term, hence 6 terms kept.

Everything in this question is that last step. The number the calculator prints is the index of the *first term small enough to drop*, and the answer is one less. The markscheme even lists the two other indexings students use — $2m + 1$ instead of $2m - 1$, or n as the exponent — and gives full marks for each, because they all end at 6.

9.2 — November 2025 TZ3 P3 Q2(h), 5 marks

The bound. The four integrals kept are of x , $-\frac{x^3}{3}$, $\frac{x^5}{5}$, $-\frac{x^7}{7}$; the first one dropped is of $\frac{x^9}{9}$:

$$\begin{aligned}\int_0^{\frac{1}{\sqrt{3}}} \frac{x^9}{9} dx &= \left[\frac{x^{10}}{90} \right]_0^{\frac{1}{\sqrt{3}}} \\ &= \frac{1}{90 \cdot 3^5} \\ &= \frac{1}{21870} \\ &= 4.57 \times 10^{-5}.\end{aligned}$$

The actual error. The exact value is $\frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3} = 0.158458558\dots$, and the approximation is 0.158422, so

$$\begin{aligned}\text{error} &= 0.158458558\dots - 0.158422 \\ &= 3.69 \times 10^{-5}.\end{aligned}$$

The comparison. $3.69 \times 10^{-5} < 4.57 \times 10^{-5}$, so the theorem holds. **AG**

Two significant figures is what the cell asks for, and deliberately: using the *unrounded* value from part (f) instead of 0.158422 gives 3.73×10^{-5} , and the markscheme accepts both. They differ in

the third figure and agree in the first two.

The **R1** is the comparison itself. Producing both numbers and not writing the inequality loses it.

9.3 — November 2025 TZ3 P3 Q2(i), 5 marks

Now the alternating terms are integrals. Term number m is

$$\begin{aligned}\int_0^{\frac{1}{\sqrt{3}}} \frac{x^{2m-1}}{2m-1} dx &= \frac{1}{2m(2m-1)} \left(\frac{1}{\sqrt{3}} \right)^{2m} \\ &= \frac{1}{2m(2m-1) 3^m}.\end{aligned}$$

Require this to be at most 1×10^{-6} :

$$m = 6 : 1.04 \times 10^{-5},$$

$$m = 7 : 2.51 \times 10^{-6},$$

$$m = 8 : 6.35 \times 10^{-7}.$$

The first term small enough to drop is $m = 8$, so **7 non-zero terms** are needed.

The markscheme's route is the same with a continuous variable: solving $\frac{(1/\sqrt{3})^{n+1}}{n(n+1)} <$

10^{-6} gives $n = 14.33\dots$, so $n = 15$ — that is the *power*, x^{15} , which is the 8th non-zero term, so seven terms are kept.

Two conversions in a row, and both are off-by-one: power $\rightarrow$ term number $\rightarrow$ terms kept. It is a five-mark question and four of the marks are arithmetic; the fifth is arriving at 7 rather than 8 or 15.

Compare with 9.1: there the terms were $\frac{x^{2m-1}}{2m-1}$ and six were needed for 10^{-4} ; here the integration divides each term by one more power of $\sqrt{3}$ and by $2m$, and seven terms buy a bound a hundred times tighter. That is the same effect that made 8.4 beat 8.3.